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LAB ASSIGNMENT-03

* **PROBLEM STATEMENT:**

Write a Java Program to implement method overloading and method overriding in Java.

1) Write a class Shapes. Find the area of basic shapes using polymorphism.

2) Write a Java program to create a parent class Hillstations and three subclasses by hill station names. Subclasses extend the superclass and extend override methods famousfood() and famousfor(). Call the methods famousfood() & famousfor() by the Parent Class, i.e. Hillstations class. It should refer to the base class object and the base class method overrides the superclass method.

* **OBJECTIVE:**
  - Study method overloading
  - Study method overriding

* **THEORY:**

(1) **Method Overloading & advantages**

→ Method Overloading is a feature in Java where multiple methods have the same name but differ in:
- No. of parameters
- Type of parameters
- Order of parameters

Example of compile-time polymorphism.

ADVANTAGES:
- Improves code readability.
- Improves code reusability
- Provides flexibility
- Reduce need for multiple method names.

② Method Overriding & Rules

Method Overriding occurs when a subclass provides a specific implementation of a method already defined in its parent class.
Example of runtime polymorphism.

RULES:
- Method name must be same as parent class.
- Parameters must be same.
- Must have IS-A relationship.
- Return type must be same or covariant.
- Access modifier can't be more restrictive.

* CONCLUSION: Thus, we have successfully implemented method overloading and method overriding in Java.

* FAQ

① Can we overload java main() method?

① Yes, we can overload the main() method in Java.
You can define multiple main() methods with diff. parameters.
But JVM only calls:

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public static void main (string[] args)

Other overloaded versions must be called manually.

(2) Can we declare overloaded methods as final?

(2) Yes, overloaded methods can be declared as final. But once declared final, they cannot be overridden in subclasses.

(3) Can an overloaded method be overridden?

(3) No, overloading and overriding are diff. concepts. However, a method that is overloaded in a parent class can still be overridden in a child class (if signature matches).

(4) What is method overriding?

(4) Method overriding is when a subclass provides a new implementation of a method already defined in its parent class.
It supports runtime polymorphism.

(5) What happens if we change the arguments of overriding method?

(5) It will not overriding anymore.
→ Changing parameters = method overloading, not overriding.

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(6) Can we change the return type from Number to Integer?

(6) Yes (Covariant Return Type)
- Allowed if the return type is a class of the original return type.